

Direction Forecasting on Daily U.S. Wheat Futures

Leakage-safe ML evaluation and a preliminary long-flat overlay

Kenji Funaki

NYU VIP Forecasting Project

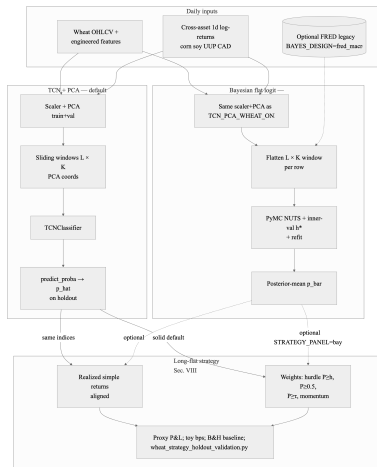
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Project objective

Forecast next-day wheat direction using public daily prices, macro data, and machine learning, then test whether predicted probabilities can support a modest tactical long-flat rule.

Data Pipeline and Leakage Controls

- Daily Chicago SRW wheat futures plus engineered technical features.
- FRED-MD macro indicators transformed by stationarity codes and shifted by one month.
- Optional cross-asset returns: corn, soybeans, UUP, and CAD.
- Chronological split: train/CV first, final 15% holdout last.
- Scalers and PCA fit only on training folds or train+validation rows.



Models and Main Forecasting Results

Model	Holdout Acc.	ROC-AUC	Macro F1
FAVAR	0.514	0.508	0.513
ARX	0.523	0.526	0.522
ELM	0.512	0.528	0.511
TCN	0.541	0.526	0.541

- TCN was the best classifier on the terminal holdout, but fold scores stayed close to 0.50.
- Reported differences are descriptive, not statistically significant superiority claims.
- Main lesson: forecasting wheat direction is hard; leakage-safe evaluation matters more than headline accuracy.

Most Recent Work: Probabilities to a Long–Flat Overlay

- Use TCN probability $\hat{p}_t$ to decide when to be long versus flat.
- Compare buy-and-hold, fixed probability hurdles, validation threshold τ , and momentum.
- Bayesian logistic MCMC adds posterior-mean probabilities $\bar{p}_t$ for uncertainty-aware variants.
- This is a frictionless proxy: no commissions, rolls, margin, options, or live execution assumptions.

Rule	Total ret.	% long
Buy-and-hold	-0.468	1.00
$p \geq 0.52$	0.000	0.00
$p \geq 0.50$	0.009	0.001
$p \geq \tau$	0.000	0.42
Momentum	-0.155	0.46

One bearish holdout tail: risk exposure mattered more than tiny accuracy gaps.

Lessons Learned and What I Would Add

- Leakage prevention is the core engineering challenge: chronological splits, fold-safe scaling, and lagged macro releases are non-negotiable.
- Alternative data and PCA helped organize features, but signal remains modest and unstable across slices.
- Nonlinear models can edge transparent baselines, but probability calibration and uncertainty are more useful than overclaiming alpha.
- A realistic next step is an expanding-window walk-forward test with nested tuning, FRED-vintage sensitivity, costs, and roll logic.
- Presentation takeaway: this is a reproducible empirical evaluation study, not an “AI trading system.”